

Model Fit

Money Ballers

2025-11-23

```
pacman::p_load(arrow, lme4, lmerTest, performance, broom, broom.mixed,  
               knitr, jtools,ggeffects, tidyverse)
```

```
options(scipen = 999)
```

```
# Set theme for plots  
theme_set(theme_minimal())
```

```
# Setting repository path  
#repo_path <- "/Users/namomac/Desktop/Moneyball_FC/inputs/"  
repo_path <- "D:/repos/Moneyball_FC/inputs/"
```

```
# Reading data  
#dat <- read_parquet(str_c(repo_path, "player_df.parquet"))  
dat <- read_parquet("player_df.parquet")
```

Table 1: Data Structure for fixed effects, except year as it is a random slope

Variable Type	Variable	Centering Strategy/ Form Factor	Justification
Time	Season Year	Min-Centering (+ Quadratic)	Establishes a meaningful starting intercept (2013 =0)
Contextual, Club	UEFA Coefficient	Year-Mean Centering	Isolates relative prestige; removes historical inflation

Variable Type	Variable	Centering Strategy/ Form Factor	Justification
Player Characteristics	Foreign Players %	Grand-Mean Centering	Moves baseline to “Average Club” (since 0% is unlikely)
	Age	Grand-Mean (+ Quadratic)	Captures absolute career stage (Prospect → Prime → Veteran)
	Height	Grand-Mean	Captures premium for absolute physical size
	Position	Reference group, attacker	Attackers tend to be valued more on average
Player Demographic	Citizen of Club Country	None (Raw binary 0/1)	Keeps “Non-Citizen” as clear reference group. Separates individual status from club composition (<code>foreign_players</code>)
Player Performance	Region of origin	Reference group, Europe	Control variable for where player is from
	Goals	Position-Centering	Preserves natural 0 and marginal value of 1 goal interpretation
	Assists	Position-Centering	Preserves natural 0 and marginal value of 1 assist interpretation
	Red Card	raw	Meaningful 0 interpretation
	Yellow Card	raw	Meaningful 0 interpretation
	Average minutes played in season	Year-Mean Centering, z-score	Isolates whether player is a starter or bench player



Figure 1: Cross-Classified Data Structure: Players move between clubs over time.

```

player_df <- dat |>
  mutate(
    player_id = factor(player_id),
    club_id   = factor(club_id),
    # reference groups
    position = fct_relevel(position, "Attack"),
    Region = fct_relevel(Region, "Europe & Central Asia"),
    # outcome on log scale
    usd_revenue_2024_log = log(usd_revenue_2024),
    # season year
    # convert season_year to 0, 1, 2, etc.
    season_year_c = season_year - min(season_year),
    season_year_c_sq = season_year_c^2, # quadratic
    # grand mean centering
    foreign_players = foreign_players - mean(foreign_players, na.rm=TRUE),
    age_c = Age - mean(Age, na.rm=TRUE),
    age_c_sq = age_c^2,
    height_c = height_in_cm - mean(height_in_cm, na.rm=TRUE)
  ) |>
  # year mean centering
  group_by(season_year_c) |>
  mutate(uefa_coefficient = uefa_coefficient - mean(uefa_coefficient, na.rm=TRUE),
         minutes_played = (minutes_played - mean(minutes_played, na.rm=TRUE)) /
           sd(minutes_played, na.rm = TRUE)) |>
  # position centering
  group_by(position) |>
  mutate(goals = goals - mean(goals, na.rm=TRUE),
         assists = assists - mean(assists, na.rm=TRUE)) |>
  ungroup()

```

- After checking missing values, we saw no more than 108 missing cases for Region, 103 for citizen of club country, and we decide to drop these cases as there are few and we are left with 7,023 cases.

```

# Which variable has missing value
player_df |>
  reframe(across(everything(), ~sum(is.na(.)))) |>
  select_if(~ . > 0) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Missing Cases") |>
  slice(1:5) |>
  arrange(desc(`Missing Cases`)) |>
  kable()

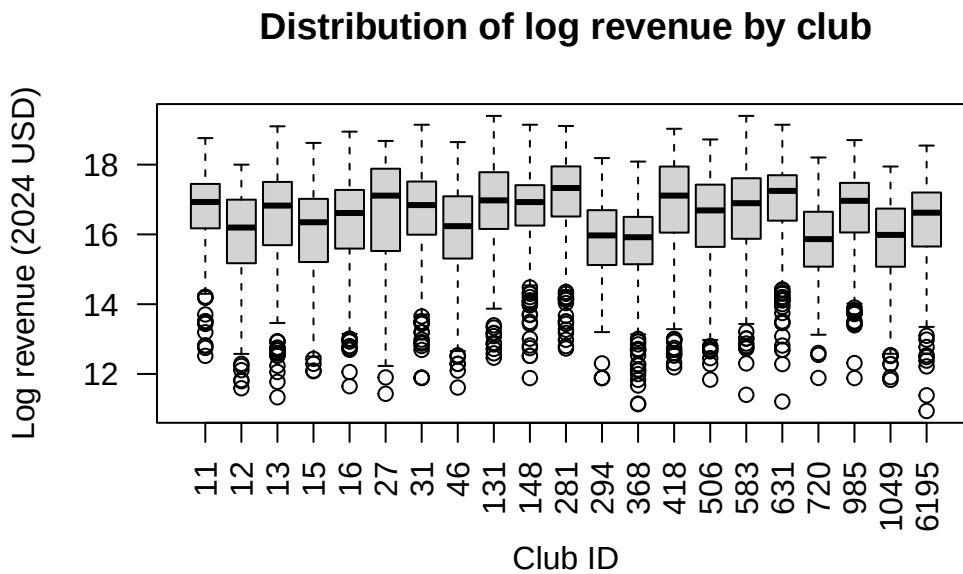
```

Variable	Missing Cases
country_of_birth	108
country_of_citizenship	103
height_in_cm	36
date_of_birth	2
Age	2

```
clean_player_df <- player_df |> drop_na()
```

- After examining the distribution of log market values across clubs, it is evident that the boxplots differ in both central tendency and variability. This indicates that club is a meaningful factor in explaining variation in market values.

```
boxplot(usd_revenue_2024_log ~ club_id,
        data = clean_player_df,
        xlab = "Club ID",
        ylab = "Log revenue (2024 USD)",
        main = "Distribution of log revenue by club",
        las = 2) # rotate x-axis labels so you can read them
```



Linearity

In the overall spaghetti plot group by `play_id`, the lines vary widely across players, showing considerable between-player heterogeneity in log market value. Within most players, the slopes appear mostly flat, but some of them change up and down dramatically.

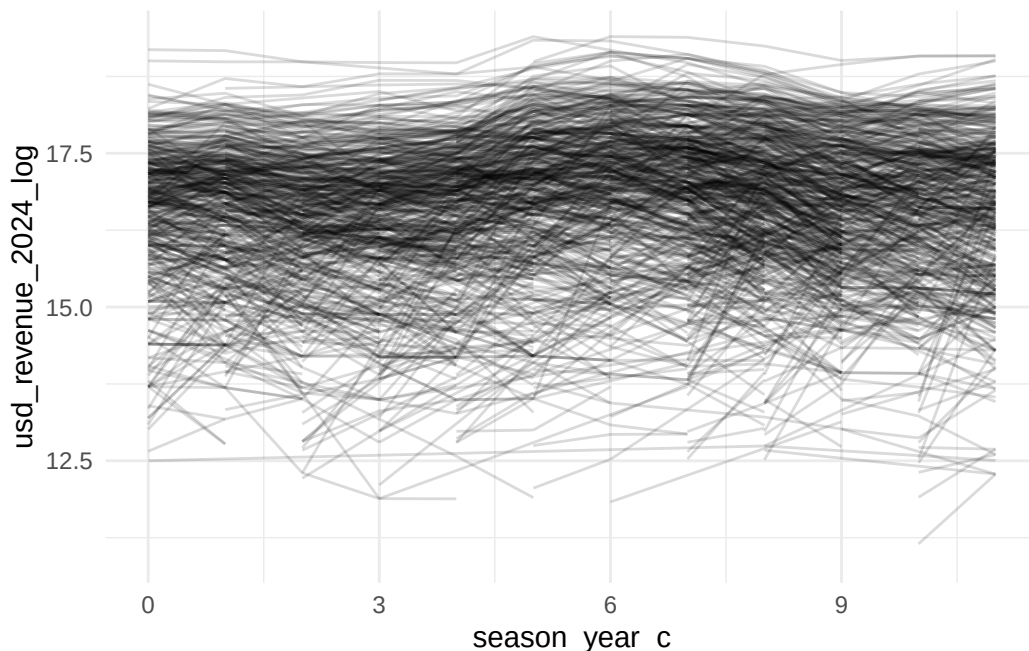
In the overall spaghetti plot group by `club_id`, they are consistently show differences. This means between-club differences in market values are large, supporting the idea that club-level random intercepts are necessary in your multilevel model.

Besides, the clubs' average values change over time, indicating slopes might vary across clubs.

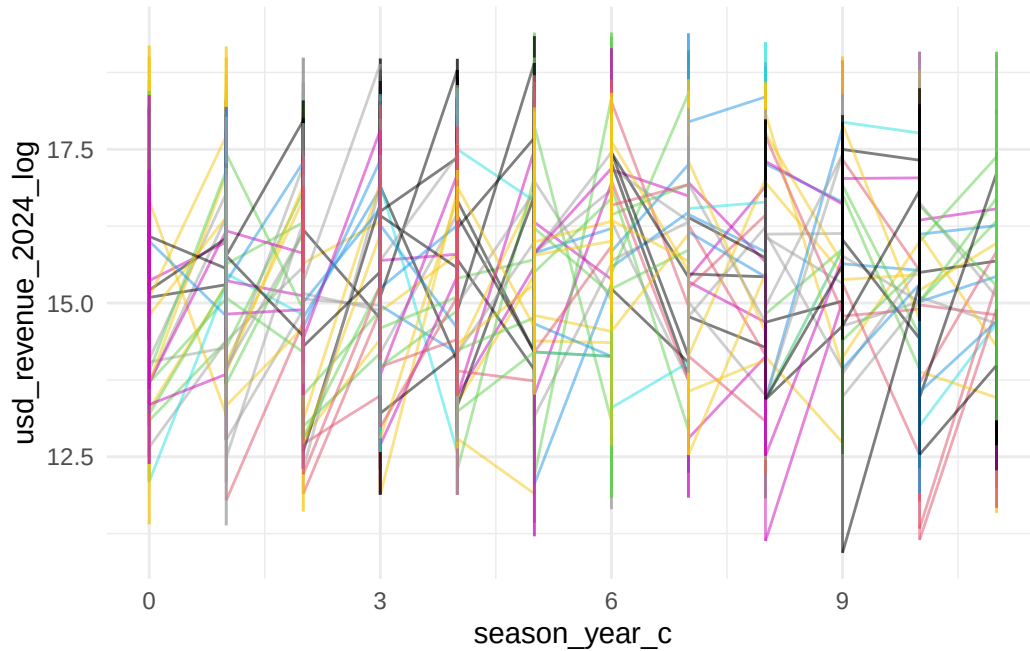
The loess smoother plot shows a non-linear averaged trend (curvature) in log-transformed market value (`usd_revenue_2024_log`) over centered season year (`season_year_c`). We can clearly see a peak at about season year 6 to 7, and the shape of the loess line is a wave pattern. Thus, we can try to add polynomial terms in our model to better explain the data.

```
library(ggplot2)

ggplot(clean_player_df) +
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log,
               group = player_id), color = "black", alpha = 0.15)
```

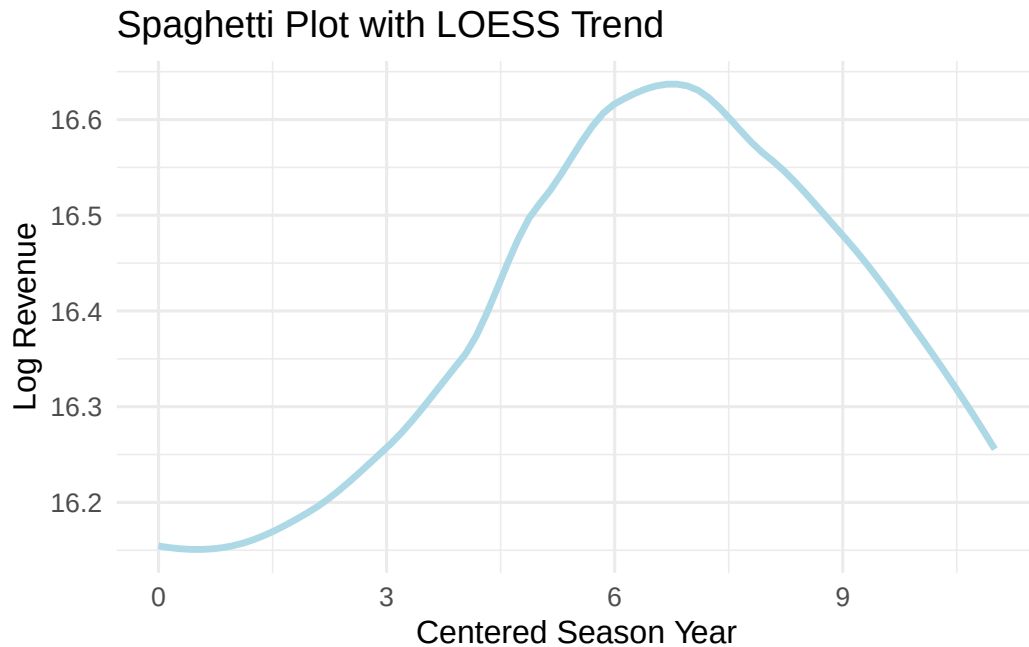


```
ggplot(clean_player_df) +
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log,
               group = club_id), color = factor(clean_player_df$club_id), alpha = 0.5)
```



```
ggplot(clean_player_df, aes(x = season_year_c,
                           y = usd_revenue_2024_log)) +

  # LOESS smoother (overall trend)
  geom_smooth(method = "loess",
             color = "lightblue", size = 1.2, se = FALSE) +
  labs(x = "Centered Season Year",
       y = "Log Revenue",
       title = "Spaghetti Plot with LOESS Trend") +
  theme_minimal(base_size = 12)
```



Modeling Process Plan

- Model 1: Random Intercepts only to examine ICC
- Testing whether a random slope on year is needed.
 - Model 2.1: Random intercepts + fixed effect of centered season__year
 - Model 2.2: Adding quadratic terms of season year
 - Model 2.3 Adding random slope of season__year_c
 - Model 2.4: Adding quadratic term of season__year as a random slope
- Adding level fixed effects
 - Establish whether age quadratic term is needed
 - * Model 3.1: Adding player covariates {Height + Position + Citizen Status + Region + Age + Goals + Assists + Red Cards + Yellow cards + Minutes played}
 - * Model 3.2: Adding the quadratic term for age

– Model 4: Adding club covariates {UEFA coefficients + Foreign Players}

- Compare model fit
- Model diagnostic
- Choose final model

Model 1, intercepts only

```
#Using bobyqa to handle the complex random effects structure and avoid gradient warnings
strict_control <- lmerControl(optimizer = "bobyqa",
                              optCtrl = list(maxfun = 200000))

Model_1 <- lmer(
  usd_revenue_2024_log ~
    # random effects
    (1|player_id) + (1|club_id), ,
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(Model_1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20105.19
BIC	20132.62
Pseudo-R ² (fixed effects)	0.00
Pseudo-R ² (total)	0.78

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.82	0.07	228.90	26.99	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.32
club_id	(Intercept)	0.28
Residual		0.72

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.75
club_id	21	0.03

- We can see that player_id accounts for about 75% of market value while club_id accounts for 3%.

Model 2.1: Adding fixed effect of year

```
Model_2.1 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c +
    # random effects
    (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(Model_2.1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20052.80
BIC	20087.09
Pseudo-R ² (fixed effects)	0.00
Pseudo-R ² (total)	0.79

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.99	0.07	221.31	32.91	0.00
season_year_c	-0.03	0.00	-8.11	6524.03	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.34
club_id	(Intercept)	0.28
Residual		0.71

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.76
club_id	21	0.03

Model 2.2 Adding quadratic term

- Notice that the linear relationship of year is no longer significant.

```
Model_2.2 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # random effects
    (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(Model_2.2)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20046.98
BIC	20088.12
Pseudo-R ² (fixed effects)	0.01
Pseudo-R ² (total)	0.79

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.92	0.07	214.88	36.55	0.00
season_year_c	0.02	0.01	1.58	5756.86	0.11
season_year_c_sq	-0.00	0.00	-4.45	5650.50	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.34
club_id	(Intercept)	0.28
Residual		0.71

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.76
club_id	21	0.03

Compare Model 2.1 and 2.2 to determine if the quadratic term is needed.

- We see that the LRT test suggests that the full model provides a significantly better fit than the reduce model, the quadratic term is needed for this data.

```
anova(Model_2.1, Model_2.2, test = "LRT")
```

Data: clean_player_df

Models:

Model_2.1: usd_revenue_2024_log ~ season_year_c + (1 | player_id) + (1 | club_id)

```

Model_2.2: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + (1 | player_id) + (1 | club_id)
      npar    AIC    BIC logLik -2*log(L) Chisq Df  Pr(>Chisq)
Model_2.1      5 20040 20074 -10015      20030
Model_2.2      6 20022 20063 -10005      20010 19.79   1 0.000008646 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Model 2.3, Random slope

- To estimate a random slope (`1 + year_centered | player_id`), the model needs multiple data points for *every* player to calculate their individual trajectory. But many players only appear in our data for 1 or 2 years, and the model struggles to distinguish the “player slope” from the “residual error,” causing these convergence issues. Clubs, however, have many players and many years of historical market value data so a random slope for clubs is more robust to fit.

```

Model_2.3 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(Model_2.3)

```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19986.28
BIC	20041.13
Pseudo-R ² (fixed effects)	0.01
Pseudo-R ² (total)	0.80

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.90	0.09	182.85	26.54	0.00
season_year_c	0.02	0.01	1.55	116.87	0.12
season_year_c_sq	-0.00	0.00	-4.66	5632.32	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.35
club_id	(Intercept)	0.35
club_id	season_year_c	0.04
Residual		0.70

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.75
club_id	21	0.05

Model 2.4, Random slope + quadratic term

- We have a convergence problem, it seems that adding the quadratic term makes the model unidentifiable. We move forward with Models 2.2 and 2.3.

```
Model_2.4 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # random effects
    (1 | player_id) + (season_year_c + season_year_c_sq | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(Model_2.4)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19961.36
BIC	20036.78
Pseudo-R ² (fixed effects)	0.01
Pseudo-R ² (total)	0.80

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.90	0.10	163.06	22.82	0.00
season_year_c	0.02	0.03	0.91	19.22	0.37
season_year_c_sq	-0.00	0.00	-2.34	19.31	0.03

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.35
club_id	(Intercept)	0.40
club_id	season_year_c	0.10
club_id	season_year_c_sq	0.01
Residual		0.70

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.74
club_id	21	0.07

Determine if the random slope on season year is needed

```
calculate_lrt_mixture <- function(reduced_model, full_model) {

  # 1. Extract Log-Likelihoods
  ll_reduced <- as.numeric(logLik(reduced_model))
  ll_full    <- as.numeric(logLik(full_model))
```

```

# 2. Calculate LRT Statistic
# Formula:  $-2 * (LL_{\text{reduced}} - LL_{\text{full}})$ 
lrt_stat <- -2 * (ll_reduced - ll_full)

# 3. Calculate Difference in Parameters (df)
# This tells us how many params we added
df_reduced <- attr(logLik(reduced_model), "df")
df_full <- attr(logLik(full_model), "df")
df_diff <- df_full - df_reduced

# 4. Calculate P-Value based on the difference
# If we added a slope + covariance, df_diff should be 2.
# The mixture is  $0.5 * \text{ChiSq}(1) + 0.5 * \text{ChiSq}(2)$ 

if (df_diff == 2) {
  p_val <- 0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE) +
    0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE)

  msg <- "Mixture of ChiSq(1) and ChiSq(2)"
} else if (df_diff == 1) {
  # If we added a parameter with NO covariance (e.g. diag structure)
  p_val <- 0.5 * pchisq(lrt_stat, df = 0, lower.tail = FALSE) +
    0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE)

  msg <- "Mixture of ChiSq(0) and ChiSq(1)"
} else {
  # Fallback for non-standard comparisons
  p_val <- pchisq(lrt_stat, df = df_diff, lower.tail = FALSE)
  msg <- paste0("Standard ChiSq(", df_diff, ")")
}

# Output
cat("LRT Statistic:", round(lrt_stat, 4), "\n")
cat("Parameter Diff:", df_diff, "\n")
cat("Distribution: ", msg, "\n")
cat("P-Value: ", format.pval(p_val, eps = .0001), "\n")
}

# Run the comparison
calculate_lrt_mixture(Model_2.2, Model_2.3)

```


LRT Statistic: 64.7002
 Parameter Diff: 2
 Distribution: Mixture of ChiSq(1) and ChiSq(2)
 P-Value: < 0.0001

- We use a helper function to print the p-value for the LRT difference, and find a p-value <.0001 and we reject the Null, (with $.5 \times 2(1) + .5 \times 2(2)$ mixture) and conclude that there is strong evidence for club variation in market value trajectories and therefore we should keep this random slope in the model.

Model 3.1: Adding fixed effects at level 1 with age as a linear relationship

- Age centered is not even significant. Same for red\yellow cards.

```
Model_3.1 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c + goals + assists + red_cards + yellow_cards + minutes_played +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(Model_3.1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	17497.18
BIC	17661.75
Pseudo-R ² (fixed effects)	0.28
Pseudo-R ² (total)	0.80

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.43	0.10	162.82	43.55	0.00
season_year_c	0.07	0.01	5.63	138.62	0.00
season_year_c_sq	-0.01	0.00	-9.05	5743.42	0.00
height_c	0.01	0.00	3.68	1677.18	0.00
positionDefender	-0.43	0.06	-7.19	1657.38	0.00
positionGoalkeeper	-1.24	0.10	-12.70	1674.56	0.00
positionMidfield	-0.22	0.06	-3.77	1685.59	0.00
citizen_of_club_countryYes	-0.40	0.04	-10.59	5702.02	0.00
RegionEast Asia & Pacific	-0.20	0.21	-0.93	1742.84	0.35
RegionLatin America & Caribbean	0.25	0.06	3.88	1803.67	0.00
RegionMiddle East & North Africa	0.10	0.25	0.39	1705.35	0.70
RegionNorth America	0.01	0.22	0.03	1787.92	0.98
RegionSub-Saharan Africa	0.02	0.10	0.21	1740.66	0.83
age_c	0.00	0.00	0.06	2366.73	0.95
goals	0.01	0.00	5.01	5556.92	0.00
assists	0.01	0.00	3.43	5095.87	0.00
red_cards	-0.01	0.03	-0.50	4713.31	0.62
yellow_cards	0.00	0.00	1.15	5428.13	0.25
minutes_played	0.52	0.02	29.58	5625.44	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.94
club_id	(Intercept)	0.37
club_id	season_year_c	0.03
Residual		0.62

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.63
club_id	21	0.10

Model 3.2. Adding age quadratic term

```
Model_3.2 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c + age_c_sq +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(Model_3.2)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14607.01
BIC	14778.43
Pseudo-R ² (fixed effects)	0.39
Pseudo-R ² (total)	0.89

- It seems that we need the quadratic term for age in the model.

```
anova(Model_3.1, Model_3.2)
```

Data: clean_player_df

Models:

Model_3.1: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + c

Model_3.2: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + c

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
Model_3.1	24	17388	17552	-8669.9	17340			
Model_3.2	25	14478	14650	-7214.1	14428	2911.8	1	< 0.000000000000000022

Model_3.1

Model_3.2 ***

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.75	0.08	216.63	81.02	0.00
season_year_c	0.10	0.01	10.59	285.23	0.00
season_year_c_sq	-0.01	0.00	-14.17	5394.55	0.00
height_c	0.01	0.00	2.64	1799.23	0.01
positionDefender	-0.40	0.06	-6.94	1785.15	0.00
positionGoalkeeper	-1.06	0.09	-11.18	1797.22	0.00
positionMidfield	-0.19	0.06	-3.24	1801.62	0.00
citizen_of_club_countryYes	-0.19	0.03	-6.05	6841.18	0.00
RegionEast Asia & Pacific	-0.45	0.20	-2.19	1836.81	0.03
RegionLatin America & Caribbean	0.19	0.06	3.09	1953.26	0.00
RegionMiddle East & North Africa	-0.08	0.24	-0.32	1814.54	0.75
RegionNorth America	-0.05	0.22	-0.23	1858.52	0.81
RegionSub-Saharan Africa	0.07	0.10	0.70	1849.10	0.49
age_c	0.02	0.00	5.43	2304.05	0.00
age_c_sq	-0.02	0.00	-61.76	6187.82	0.00
goals	0.00	0.00	1.88	5191.96	0.06
assists	0.01	0.00	3.41	4899.62	0.00
red_cards	0.01	0.02	0.64	4687.80	0.52
yellow_cards	-0.00	0.00	-0.99	5112.21	0.32
minutes_played	0.36	0.01	26.55	5267.38	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.95
club_id	(Intercept)	0.23
club_id	season_year_c	0.02
Residual		0.47

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.77
club_id	21	0.05

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- After adding cubic terms, it yields a better fit. It cannot only explain the peak of the overall trajectory, but can also explain the up and down wave pattern over time of the trajectory.

```
season_year_c_cub <- clean_player_df$season_year_c^3

Model_cub <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq + season_year_c_cub +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c + age_c_sq +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # club fixed effects
    uefa_coefficient + foreign_players +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

anova(Model_3.2, Model_cub)
```

Data: clean_player_df

Models:

Model_3.2: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + c

Model_cub: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + season_year_c_cub + hei

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
--	------	-----	-----	--------	-----------	-------	----	------------

Model_3.2	25	14478	14650	-7214.1	14428			
-----------	----	-------	-------	---------	-------	--	--	--

Model_cub	28	14362	14554	-7152.9	14306	122.24	3	< 0.00000000000000022
-----------	----	-------	-------	---------	-------	--------	---	-----------------------

Model_3.2

Model_cub ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model 4: Adding club covariates

- The model with club covariates underperforms compared to Model 3.2.

```
Model_4 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c + age_c_sq +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # club fixed effects
    uefa_coefficient + foreign_players +
    # random effects
    (1 | player_id) + (season_year_c | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(Model_4)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14622.02
BIC	14807.16
Pseudo-R ² (fixed effects)	0.39
Pseudo-R ² (total)	0.89

```
anova(Model_3.2, Model_4)
```

Data: clean_player_df

Models:

Model_3.2: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + c

Model_4: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + cit

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
Model_3.2	25	14478	14650	-7214.1	14428			
Model_4	27	14479	14664	-7212.4	14425	3.3897	2	0.1836

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.76	0.08	214.00	76.65	0.00
season_year_c	0.10	0.01	10.00	274.96	0.00
season_year_c_sq	-0.01	0.00	-13.37	5374.35	0.00
height_c	0.01	0.00	2.65	1796.69	0.01
positionDefender	-0.40	0.06	-6.95	1782.40	0.00
positionGoalkeeper	-1.06	0.09	-11.18	1794.71	0.00
positionMidfield	-0.19	0.06	-3.24	1798.93	0.00
citizen_of_club_countryYes	-0.19	0.03	-5.95	6851.81	0.00
RegionEast Asia & Pacific	-0.45	0.20	-2.19	1834.38	0.03
RegionLatin America & Caribbean	0.19	0.06	3.07	1951.10	0.00
RegionMiddle East & North Africa	-0.08	0.24	-0.33	1811.89	0.74
RegionNorth America	-0.05	0.22	-0.24	1856.11	0.81
RegionSub-Saharan Africa	0.07	0.10	0.68	1846.84	0.50
age_c	0.02	0.00	5.41	2303.32	0.00
age_c_sq	-0.02	0.00	-61.75	6198.22	0.00
goals	0.00	0.00	1.90	5187.13	0.06
assists	0.01	0.00	3.42	4895.10	0.00
red_cards	0.01	0.02	0.61	4682.99	0.54
yellow_cards	-0.00	0.00	-0.99	5107.27	0.32
minutes_played	0.36	0.01	26.56	5262.54	0.00
uefa_coefficient	-0.00	0.00	-1.16	4432.07	0.24
foreign_players	0.14	0.10	1.46	1851.74	0.14

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.95
club_id	(Intercept)	0.24
club_id	season_year_c	0.02
Residual		0.47

Bonus round, Model 5 testing 2 levels only for player data.

- In a two level model dropping the random intercept on club, which ultimately drops the random slope, is a worst fit than model 3.2. It appears we need the random slope of year, but not necessarily the the club covariates.

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.77
club_id	21	0.05

```
Model_5 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c + season_year_c_sq +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c + age_c_sq +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # random effects
    (1 | player_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(Model_5)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14928.80
BIC	15079.65
Pseudo-R ² (fixed effects)	0.38
Pseudo-R ² (total)	0.89

```
anova(Model_3.2, Model_5)
```

Data: clean_player_df

Models:

Model_5: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + cit.

Model_3.2: usd_revenue_2024_log ~ season_year_c + season_year_c_sq + height_c + position + c.

	npair	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
Model_5	22	14798	14949	-7377.1	14754			
Model_3.2	25	14478	14650	-7214.1	14428	326.17	3	< 0.00000000000000022

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.78	0.06	275.62	2451.14	0.00
season_year_c	0.10	0.01	11.22	6862.95	0.00
season_year_c_sq	-0.01	0.00	-13.77	5394.07	0.00
height_c	0.01	0.00	2.29	1899.04	0.02
positionDefender	-0.37	0.06	-5.98	1902.41	0.00
positionGoalkeeper	-1.03	0.10	-10.25	1911.41	0.00
positionMidfield	-0.15	0.06	-2.42	1917.48	0.02
citizen_of_club_countryYes	-0.27	0.03	-8.59	7002.32	0.00
RegionEast Asia & Pacific	-0.48	0.22	-2.23	1942.91	0.03
RegionLatin America & Caribbean	0.09	0.06	1.41	2016.74	0.16
RegionMiddle East & North Africa	-0.16	0.25	-0.65	1929.81	0.51
RegionNorth America	-0.10	0.23	-0.42	1970.98	0.68
RegionSub-Saharan Africa	0.03	0.11	0.26	1949.84	0.79
age_c	0.02	0.00	5.04	2366.63	0.00
age_c_sq	-0.02	0.00	-63.69	6042.27	0.00
goals	0.00	0.00	1.59	5197.50	0.11
assists	0.01	0.00	3.47	4983.99	0.00
red_cards	-0.00	0.02	-0.04	4796.43	0.97
yellow_cards	-0.01	0.00	-2.91	5182.20	0.00
minutes_played	0.36	0.01	26.02	5278.50	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.02
Residual		0.47

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.82

Model_5

Model_3.2 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
export_summs(Model_2.3, Model_3.2, Model_4,
             digits=3, r.squared=FALSE,
             model.names = c("Model_2.3", "Model_3.2", "Model_4"))
```

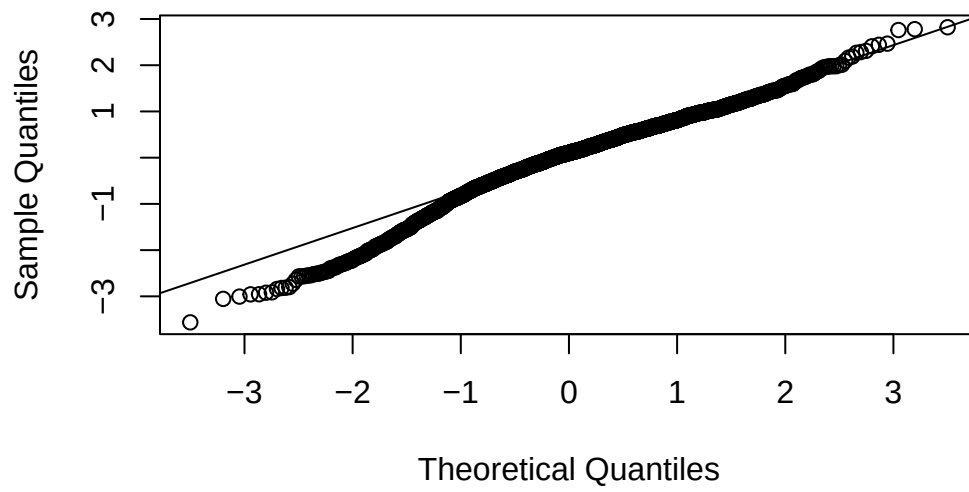
```
final_model <- Model_3.2
```

Model Diagnostics

- By comparing the QQ-line and QQ-plot (both residuals and random effects) we can find that both model plots look good on normality, except some outliers. By comparing the residual-fitted value plot, we can find that both model show the same spread pattern, the dots are evenly spread across zero-mean line up and down.
- By comparing EBLUPs plots, we can see that most of the points lie deviate from the diagonal at both tails, and the lower side is more downward, suggesting moderate departures from normality in the random slopes and random intercepts, which indicates possible skewness and outliers. Thus, we cannot say the normal random effects assumption in the multilevel model are met.

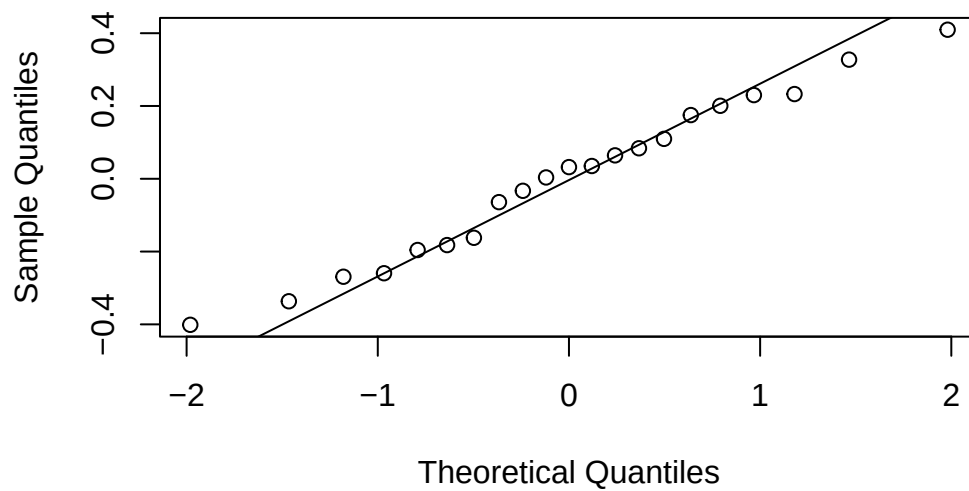
```
# slopes
qqnorm(ranef(final_model)$player_id[,1]) # group factor
qqline(ranef(final_model)$player_id[,1])
```

Normal Q-Q Plot

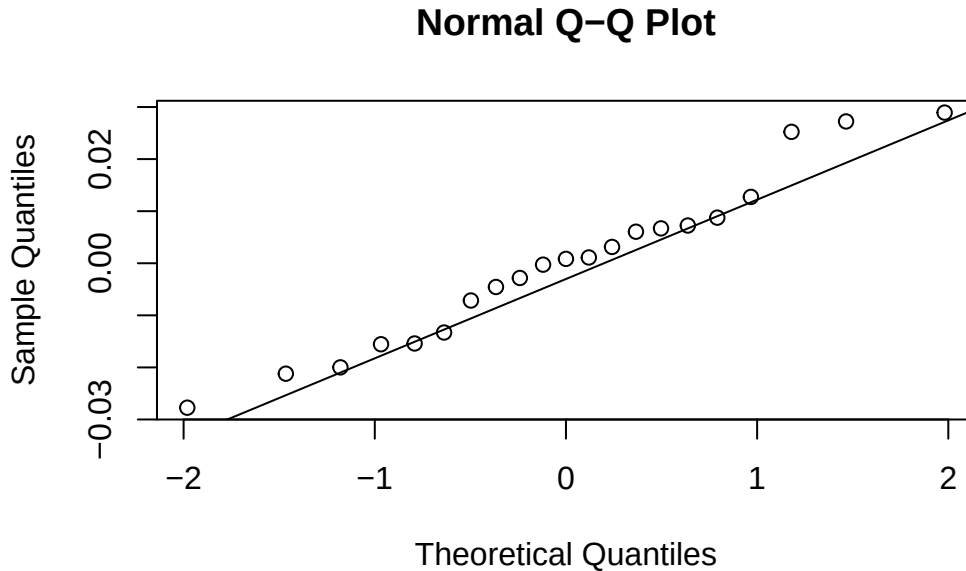


```
qqnorm(ranef(final_model)$club_id[,1]) # group factor  
qqline(ranef(final_model)$club_id[,1])
```

Normal Q-Q Plot



```
qqnorm(ranef(final_model)$club_id[,2]) # group factor
qqline(ranef(final_model)$club_id[,2])
```



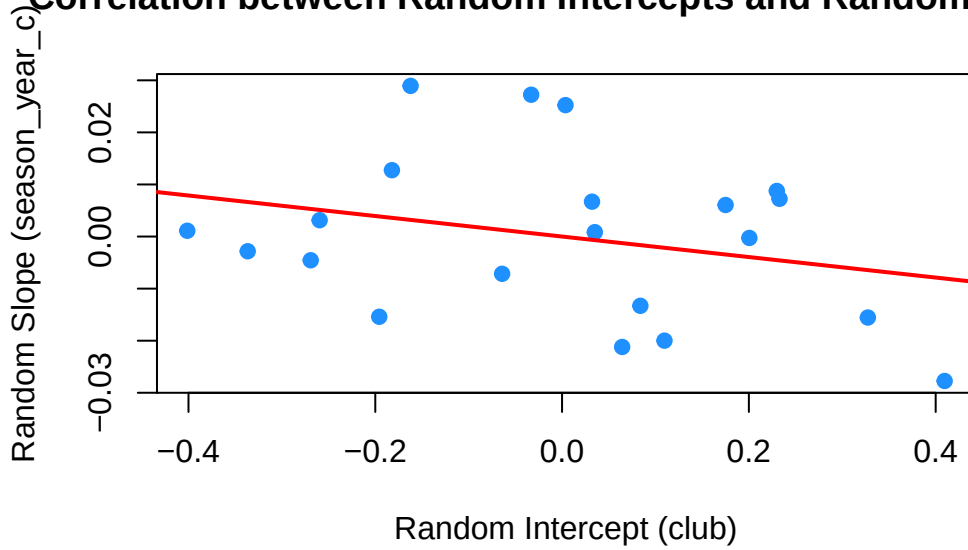
- The red regression line slopes slightly downward, but only gently. That is, clubs with higher baseline log market value tend to have slightly more negative or flatter slopes. Clubs with lower baseline market value tend to have slightly steeper slopes.
- Most residuals lie between about -1 and $+1$, and fitted log revenues between 14 and 18. There is no strong curved pattern or obvious systematic relationship between residual and fitted values. So the residual reasonably consistent with the linearity assumption. Besides, there is no funnel shape, so the residuals are homogeneous. However, we can still see some outliers exist.

```
re <- ranef(final_model)$club_id

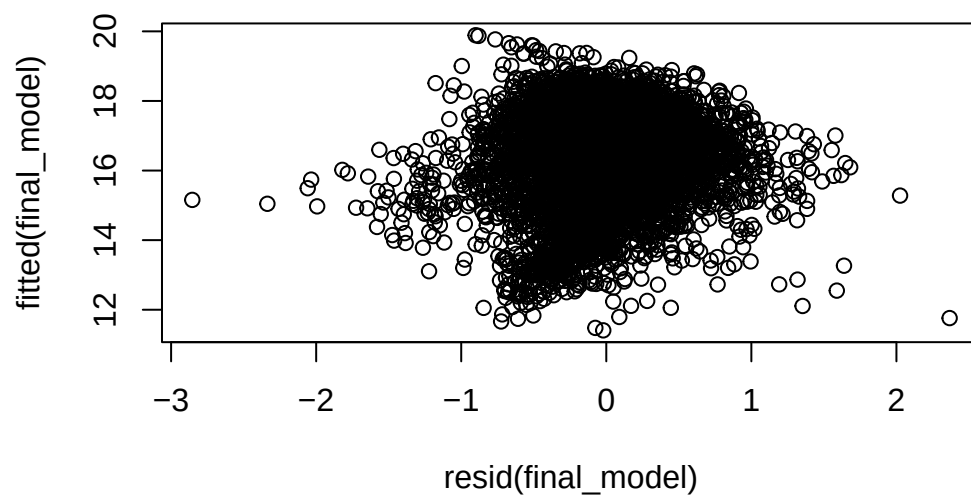
plot(
  re[, "(Intercept)"],
  re[, "season_year_c"],
  xlab = "Random Intercept (club)",
  ylab = "Random Slope (season_year_c)",
  main = "Correlation between Random Intercepts and Random Slopes",
  pch = 19, col = "dodgerblue"
)
```

```
abline(lm(re[, "season_year_c"] ~ re[, "(Intercept)"]), col = "red", lwd = 2)
```

Correlation between Random Intercepts and Random Slope



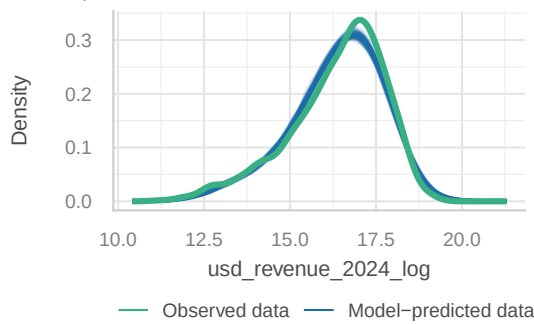
```
# Residual scatterplot  
plot(resid(final_model), fitted(final_model))
```



```
check_model(final_model)
```

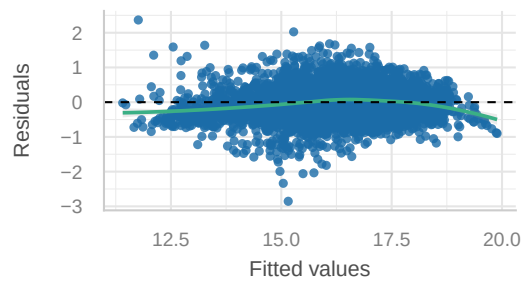
Posterior Predictive Check

Model-predicted lines should resemble observed data line



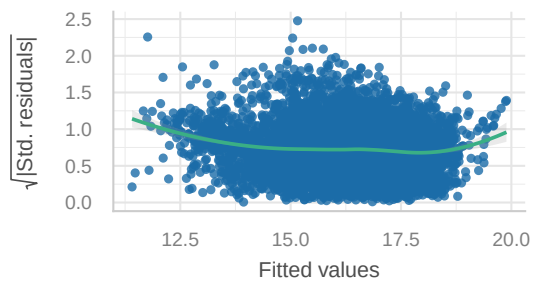
Linearity

Reference line should be flat and horizontal



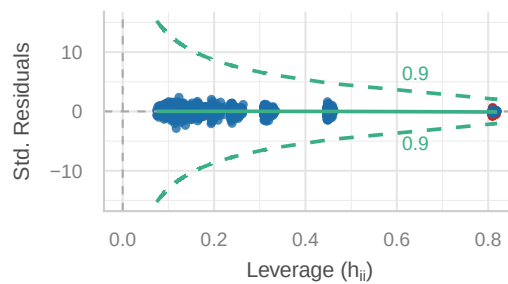
Homogeneity of Variance

Reference line should be flat and horizontal



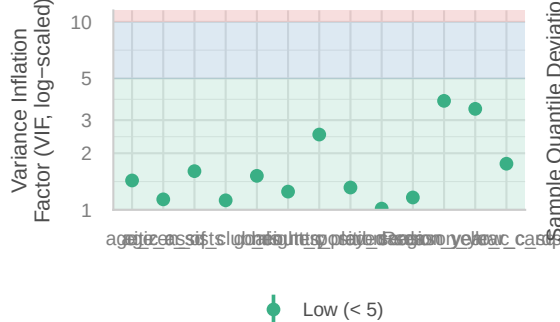
Influential Observations

Points should be inside the contour lines



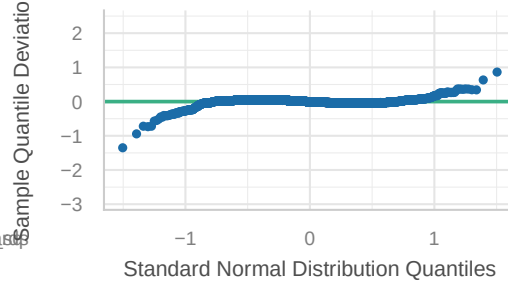
Collinearity

High collinearity (VIF) may inflate parameter uncertainty



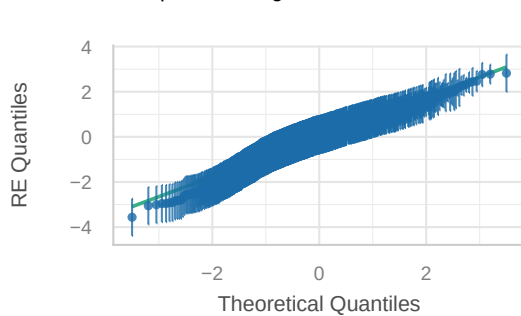
Normality of Residuals

Dots should fall along the line



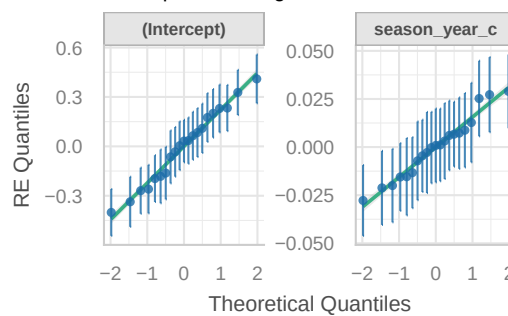
Normality of Random Effects (player_id)

Dots should be plotted along the line



Normality of Random Effects (club_id)

Dots should be plotted along the line



Final Model Specification

To test our hypotheses regarding the determinants of player market value, we fitted a **cross-classified multilevel model** (Model 3.2). This structure accounts for the non-hierarchical nature of the data, where players (i) move between different clubs (j) over time (t).

The final model specification for the log-transformed market value, Y_{tij} , is formally expressed as:

$$\begin{aligned}
 \log(\text{Revenue})_{tij} = & \beta_0 + \underbrace{\beta_1(\text{Year}_t) + \beta_2(\text{Year}_t^2)}_{\text{Time Trends}} \\
 & + \underbrace{\beta_3(\text{Height}_i - \bar{H}) + \beta_4(\text{Citizen}_{ij}) + \beta_5(\text{Age}_{it} - \bar{A}) + \beta_6(\text{Age}_{it}^2)}_{\text{Player Fixed Effects (Traits \& Demographics)}} \\
 & + \underbrace{\sum_p \beta_{Pos,p}(\mathbb{I}_{\text{Position}_i=p}) + \sum_r \beta_{Reg,r}(\mathbb{I}_{\text{Region}_i=r})}_{\text{Categorical Controls}} \\
 & + \underbrace{\beta_7\text{Goals}_{it} + \beta_8\text{Assists}_{it} + \beta_9\text{Red}_{it} + \beta_{10}\text{Yellow}_{it} + \beta_{11}(\text{Mins}_{Z,it})}_{\text{Performance \& Status (Time-Varying)}} \\
 & + \underbrace{u_{0i}}_{\text{Player Random Intercept}} \\
 & + \underbrace{v_{0j} + v_{1j}(\text{Year}_t)}_{\text{Club Random Effects}} \\
 & + \epsilon_{tij}
 \end{aligned}$$

Variance-Covariance Structure:

$$\begin{aligned}
 u_{0i} & \sim N(0, \sigma_{\text{player}}^2) \\
 \begin{pmatrix} v_{0j} \\ v_{1j} \end{pmatrix} & \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{\text{club_int}}^2 & \sigma_{\text{cov}} \\ \sigma_{\text{cov}} & \sigma_{\text{club_slope}}^2 \end{pmatrix} \right) \\
 \epsilon_{tij} & \sim N(0, \sigma_{\epsilon}^2)
 \end{aligned}$$

Interpretation of Results

Fixed Effects (Average Predictors)

- **Intercept** ($\beta_0 = 16.75$): The estimated log market value for a baseline player (Attacker from Europe/Central Asia) with average physical traits and career status at the start of the study period (2013).

- **Market Trajectory** ($\beta_1 = 0.10, \beta_2 = -0.01$): The significant positive linear term indicates the transfer market grew substantially over the decade. However, the significant negative quadratic term (β_2) reveals a **market deceleration**. The peak of this quadratic curve occurs roughly at Year 5 (2018), aligning with the historical “post-Neymar” boom followed by a correction period.
- **Player Status** ($\beta_{11} = 0.36$): The variable `minutes_played` was standardized (Z-scored) to improve model convergence. The positive coefficient indicates that a **one standard deviation increase** in playing time is associated with a 36% increase in market value, holding all else constant. This confirms that active “Squad Status” is a primary driver of valuation.
- **Biological Age** ($\beta_5 = 0.02, \beta_6 = -0.02$): Using grand-mean centering, we observe a classic parabolic career arc. Value increases as a player moves from “Prospect” to “Prime,” but the negative quadratic term confirms that value declines significantly as players enter their “Veteran” years.
- **Homegrown Effect** ($\beta_4 = -0.19$): The variable `citizen_of_club_country` serves as a proxy for domestic status. The negative coefficient indicates that citizens are valued lower than their non-citizen counterparts. Converting from the log scale ($e^{-0.19} - 1 \approx -0.173$), we find that **domestic players are valued approximately 17.3% lower** than foreign players, likely reflecting the premium paid to import talent from abroad.
- **Physical Traits** ($\beta_3 = 0.01$): We observe a small but significant **Physical Premium**. For every 1cm a player is taller than the global average, their market value increases by approximately 1%, suggesting the market favors physical stature.

Random Effects (Variance Components)

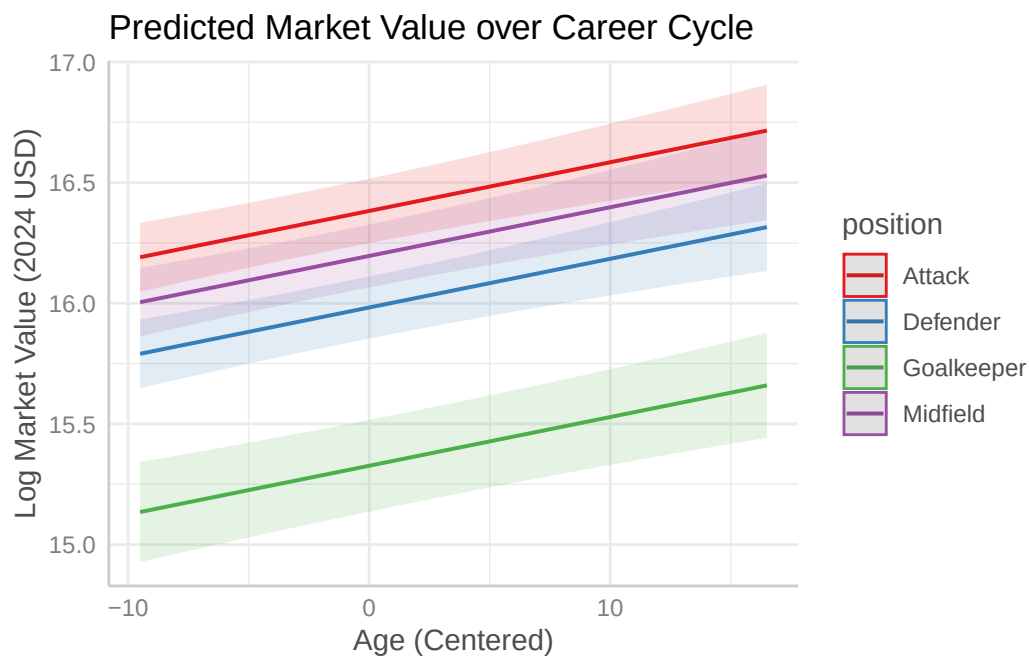
- **Player Heterogeneity** ($\sigma_{player} = 0.95$): The substantial random intercept for players (ICC ≈ 0.77) suggests that **reputation and brand** (“The Superstar Effect”) drive the majority of market value variance, rather than just annual performance statistics.
- **Club Trajectories** ($\sigma_{slope} = 0.02$): The significant random slope for `season_year` confirms that clubs do not share a parallel growth path. Some clubs act as “Incubators” (accelerating player value faster than the market), while others remain stagnant.
- **The “Catch-Up” Effect** ($r = -0.35$): The negative correlation between the Club Intercept and Club Slope suggests that clubs starting with lower baseline values (smaller clubs) often see steeper relative growth rates in their assets compared to established super-clubs.

Visualization of Marginal Effects

To better visualize the non-linear effect of Age on Market Value, we computed marginal effects holding all other predictors at their mean.

```
# Visualize the Quadratic Age Effect
# We select representative ages (18 to 38) to show the curve
age_pred <- ggpredict(Model_3.2, terms = c("age_c [all]", "position"))

plot(age_pred) +
  labs(
    title = "Predicted Market Value over Career Cycle",
    y = "Log Market Value (2024 USD)",
    x = "Age (Centered)"
  )
```



Limitations

While our model demonstrates strong predictive power (R^2 0.89), several limitations warrant mention:

1. **Small Cluster Size for Random Slopes:** We estimated random slopes for `Club`, but our dataset contains only 21 unique clubs. While the Likelihood Ratio Test supported the inclusion of this slope ($p < .001$), variance estimates from small numbers of clusters should be interpreted with caution.
2. **Normality of Residuals:** Diagnostic plots (QQ plots) indicated deviation from normality in the extreme tails of the distribution. While the large sample size ($N=7023$) ensures robust estimation of fixed effects, future iterations could employ robust standard errors or a Gamma GLMM to better handle these extremes.
3. **Endogeneity:** Predictors such as `minutes_played` are likely endogenous (better players get more minutes). The coefficient for this variable should be interpreted as an association reflecting “Squad Status” rather than a purely causal effect of playing time.

	Model_2.3	Model_3.2	Model_4
(Intercept)	15.903 *** (0.087)	16.752 *** (0.077)	16.757 *** (0.078)
season_year_c	0.022 (0.015)	0.099 *** (0.009)	0.096 *** (0.010)
season_year_c_sq	-0.005 *** (0.001)	-0.010 *** (0.001)	-0.009 *** (0.001)
height_c		0.010 ** (0.004)	0.010 ** (0.004)
positionDefender		-0.400 *** (0.058)	-0.400 *** (0.058)
positionGoalkeeper		-1.056 *** (0.094)	-1.055 *** (0.094)
positionMidfield		-0.186 ** (0.058)	-0.186 ** (0.057)
citizen_of_club_countryYes		-0.192 *** (0.032)	-0.189 *** (0.032)
RegionEast Asia & Pacific		-0.446 * (0.203)	-0.446 * (0.203)
RegionLatin America & Caribbean		0.188 ** (0.061)	0.186 ** (0.061)
RegionMiddle East & North Africa		-0.076 (0.237)	-0.077 (0.237)
RegionNorth America		-0.051 (0.216)	-0.052 (0.216)
RegionSub-Saharan Africa		0.069 (0.100)	0.068 (0.100)
age_c	36	0.020 *** (0.004)	0.020 *** (0.004)
age_c_sq		-0.024 *** (0.000)	-0.024 *** (0.000)
goals		0.004	0.004